

AviaGames

Duke Product Lab

Insights Report

From Consumer Discovery to Product Strategy: Identifying Growth Opportunities in Cognitively Motivated Player Segments.

Table of Contents

Table of Contents	1
Executive Summary	2
Problem Statement	3
Literature Review	4
Competitive Analysis	5
Lumosity Case Study	7
Discussion of the Pivot	8
SWOT Analysis	9
Customer Discovery	10
User Segmentation	15
User Interviews	17
Conclusions, Recommendations, and Next Steps	18

Executive Summary

This report presents findings from two sprints of consumer discovery conducted in partnership with AviaGames through Duke University's Product Lab.

Overview and Objectives

This report presents findings from two sprints of consumer discovery conducted in partnership with AviaGames through Duke's Product Lab. The initial objective was to explore whether Avia could expand into the "brain training" space, driven by user perceptions that gameplay may support memory, focus, or cognitive engagement.

Scope and Limitations

Rather than assuming this opportunity existed, we grounded our approach in a structured evaluation of the problem space. Through a combination of literature review, competitive analysis, and case study examination, we identified a critical distinction between perceived cognitive benefits and clinically validated outcomes—revealing both meaningful opportunity and significant risk in how products in this category are positioned.

Strategic Reframe

Given these findings, we reframed the problem from ***"Can Avia enter brain training?"***

to

"How might Avia responsibly capture value from cognitively motivated players?"

Customer Discovery and Validation

To validate this opportunity, we conducted large-scale consumer discovery, analyzing over 1,000 survey responses and segmenting users based on their underlying motivations for play. Our findings highlight the presence of a distinct group of players who associate gameplay with cognitive engagement, despite the absence of medical validation

What this report delivers

This report synthesizes these insights to establish a consumer-driven foundation for product strategy. It outlines how Avia can leverage player motivations, navigate positioning risks, and explore growth opportunities—while remaining grounded in responsible and defensible product decisions.

Problem Statement

How can we best assess player perceptions of gameplay in order to position Avia's products accordingly while fostering a beneficial research partnership between Avia and Duke?

Avia partnered with Duke Product Lab after collecting anecdotal evidence from users about the cognitive benefits of playing their games, particularly Solitaire Clash. Our team was tasked with laying the foundation for research focused on assessing the validity of these player perceptions and determining how to appropriately cater to them.

We have the unique opportunity to interface with Avia's player base in addition to collaborating with Professors **Enesto Escobar** and **Brian Chu** from the Game Development Department at Duke University through this research initiative.

Key Questions for Players

1. Why do you feel that gameplay improves cognition?
 2. What does improved cognition look like to you?
 3. What aspects of your life have felt improved after you started playing Avia's games?
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Reality Check: Gaming vs Cognition

Research over the past two decades has increasingly examined whether cognitively stimulating activities—including digital games, board games, and structured cognitive training—can enhance cognitive functioning or delay cognitive decline in older adults. Overall, the literature suggests promising but nuanced findings.

Literature Review

Randomized controlled trials (RCTs) provide some of the strongest evidence in this area. For example, Feng Lin et al. (2016) conducted a randomized controlled pilot trial examining vision-based speed-of-processing training in older adults with mild cognitive impairment. The study found improvements not only in trained domains (processing speed and attention) but also transfer effects to untrained domains such as working memory and instrumental activities of daily living . Neural changes in resting-state networks were also observed, suggesting possible neuroplastic effects.

Similarly, a study protocol by Xiaojuan Shi et al. (2023) describes a cluster randomized controlled trial investigating chess and card-based leisure activities as a non-pharmacological intervention for cognitive function in adults over 60 . The rationale is grounded in the cognitive reserve hypothesis: intellectually stimulating activities may strengthen neural networks and delay cognitive decline.

A systematic review by Aizuka et al. (2019) examined 20 intervention studies using cognitive leisure activities (e.g., reading, writing, board games, handicrafts). Thirteen studies reported improvements in at least one cognitive domain, particularly working memory and global cognition . However, effects were often broad rather than domain-specific, and methodological quality varied.

Large-scale population studies further the association between cognitively engaging leisure activities and better cognitive performance. For example, Du et al. (2025) found that participation in mahjong, chess, and card games was significantly associated with higher cognitive scores among middle-aged and older adults in China . However, as a cross-sectional study, causal inference cannot be established.

Across studies, recurring mechanisms proposed include:

- Cognitive reserve enhancement
- Neuroplastic adaptation
- Increased attention, working memory, and executive load
- Social engagement as a mediating factor

Despite encouraging findings, the literature also highlights significant methodological challenges. Many studies rely on:

- Small sample sizes
- Short intervention periods
- Limited long-term follow-up
- Variable control conditions
- Domain-specific rather than ecologically valid outcomes

Importantly, demonstrating cognitive enhancement requires:

- Large-scale randomized controlled trials
- Active control groups
- Blinded assessments
- Longitudinal follow-up
- Clear evidence of transfer to real-world functioning

Such requirements typically exceed the scope of exploratory product-lab collaborations and require standalone clinical research infrastructure.

Competitive Analysis

Avia Games is positioned as a mobile-first skill-based competitive gaming company, focusing on accessible yet structured gaming experiences across multiple genres. Its competitive advantages lie in a unified platform, strong player engagement, and a curated game portfolio. However, it remains in a competitive landscape with both established platforms (e.g., Skillz) and emerging challengers (e.g., Z League). Legal history and market positioning challenges reinforce the need for differentiation through innovation, UX excellence, and compliance focus.

Z League

Feature	Z League	Avia Games
Primary focus	PvP tournaments	Skill-based mobile games & platform
Platform	Mobile/web	Mobile first + unified app ecosystem
Monetization	Tournaments & rewards	Skill-based play & real-money matches
Reach	Smaller niche	Larger user base & multiple game types

Skillz

Feature	Skillz	Avia Games
Platform	Broad tournament ecosystem	Specialized curated games
Scale	Larger portfolio, publicly traded	Growth stage (Series B)
Competitive history	Patent litigation victory	Emerging competitor
Monetization	Real-money contests	Real-money + freemium formats

OkLetsPlay

Feature	OkLetsPlay	Avia Games
Focus	Tournament platform	Game developer & platform
Platform	Cross-platform support	Primarily mobile
Player reach	Smaller, niche	Larger player base & marketing presence

Miscellaneous Competitors

Other players include Triumph Labs (monetization focus), Blast, Game Taco, and Sachar Gaming — each contributing in parts to the skill-based, competitive, or monetization layers of the mobile games ecosystem.

Competitive Analysis

Market Positioning & Strategic Differentiators

- **Large player engagement:** 60M+ downloads, 450M+ monthly tournaments, and strong category rankings on app stores indicate broad adoption.
- **Unified platform experience:** Pocket7Games connects multiple genres - from card and puzzle to sports and bingo - under one membership.
- **Brand & industry recognition:** Awards and nominations (2025 TITAN Business Award, a 2025 Bronze Stevie Award for customer service, and multiple 2024 NYX Game Awards) at industry events reinforce credibility.

- **Legal & perception risk:** Past litigation with competitors (e.g., patent disputes with Skillz) introduces reputational and IP risk.
- **Niche competitive segment:** Skill-based real-money gaming sits between casual gaming and esports, which narrows mainstream reach relative to major mobile publishers.

- **Growing demand for mobile competitive play:** The success of structured, skill-based games highlights a trend toward short competitive formats.
- **Partnerships & collaborations:** Events like GDC and partnerships with payment providers help Avia extend reach and professional credibility.

- **Other platforms scaling competitive ecosystems:** Larger platforms like Skillz or multi-genre publishers could attract players or developers.
- **Regulation and compliance:** Real-money competitions in mobile gaming often attract scrutiny and require careful jurisdictional compliance.

Case Study: Lumosity Lawsuit

In 2016, the U.S. Federal Trade Commission (FTC) took enforcement action against Lumos Labs, the company behind the commercial “brain-training” program Lumosity, by filing deceptive advertising charges and securing a settlement worth approximately \$2 million. This case is instructive for understanding the legal and evidentiary standards that apply when products make claims about cognitive enhancement.

What Lumosity Claimed and How It Was Marketed

Lumosity marketed a suite of brain games as a structured cognitive training program designed to improve memory, attention, processing speed, and other cognitive functions. Its advertising messaging suggested that playing the games could:

- Improve performance on everyday tasks such as work, school, or athletics
- Protect against age-related cognitive decline, including mild cognitive impairment and dementia
- Reduce cognitive effects of serious health conditions, including stroke, PTSD, ADHD, and the side effects of chemotherapy

Why the FTC Challenged the Advertising

The FTC alleged that Lumosity’s advertising claims lacked the required “competent and reliable scientific evidence,” as the company did not have sufficient clinical research demonstrating that its games led to measurable improvements in real-world cognitive functioning.

Under the resulting settlement order:

- Lumos Labs agreed to pay \$2 million in redress to consumers.
- It is prohibited from making any representation that its products provide cognitive, functional, or health benefits unless backed by high-quality human clinical testing that is randomized, controlled, and sufficiently blinded, conducted by qualified researchers.
- The company must notify subscribers about the FTC action and allow them to cancel auto-renewal plans.

Implications for Cognitive Claims in Advertising

The Lumosity case highlights an important regulatory standard: marketing claims that a product will improve cognitive performance or protect against cognitive decline cannot be made on the basis of correlational or exploratory evidence alone. They require rigorous, causal evidence that shows real-world effects beyond mere improvements in performance on trained tasks.

Why This Matters for Our Project

Although the literature review for this collaboration identified associations between game-like tasks and cognitive outcomes, including some promising findings on attention, memory, and processing speed, this body of research does not meet the evidentiary threshold that courts and regulators require for public claims of cognitive enhancement. Most existing studies are limited by design issues such as small samples, lack of active control groups, short training periods, and narrow outcome measures. Furthermore, transfer effects — improvement on untrained real-world tasks — remain difficult to demonstrate convincingly in many cases.

Our Pivot

After validating our direction with an expert in cognitive aging, we reassessed our approach.

What we found

- *Limited causal evidence linking gaming to improved cognition*
- *Regulatory and legal risks around “brain training” claims*
- Misalignment with our scope, requiring a strategic shift

So we pivoted to...

- *Understanding why users play (motivation > medical claims)*
- Analyzing perceived vs. actual benefits
- Designing experiences around user-driven value, not clinical outcomes

SWOT Analysis

With this pivot in mind, we've done a SWOT Analysis of Avia to see where it stands.

STRENGTHS • Large and engaged user base • Different game offerings providing a variety of user segments • Strong brand presence providing credibility	WEAKNESSES • Currently gathering strictly anecdotal evidence to address of a medical question • Lack of strong evidence supporting generalizable cognitive benefits of gameplay in literature
OPPORTUNITIES • Potential for testing at a small and large scale with user base • Access to subject matter experts in relevant fields to pursue problem statement • Ability to further refine games for different user segments	THREATS • Other (mobile) games that may also appeal to cognitive stimulation segment • Advertisement regulation and compliance concerns

Customer Discovery: Data and Methodology

Sample selection

To evaluate player motivations and identify potential growth opportunities, we designed and distributed a structured survey to AviaGames players, resulting in over 1,000 user responses. The survey was intentionally built to capture how users engage with gameplay and the value they associate with it—particularly around perceptions of cognitive engagement such as memory, focus, and mental stimulation.

In addition to quantitative responses, the survey included an opt-in component for follow-up research. This allowed us to recruit participants for user interviews, ensuring that our qualitative insights were grounded in real player experiences and directly tied to observed patterns in the data. Our sampling approach prioritized capturing a diverse range of player motivations, with particular attention to users who associate gameplay with cognitive engagement.

✦ At a Glance

- 1,000+ survey responses
- Mixed-methods approach
- Opt-in interviews conducted
- Focus on player motivation segmentation

Data analysis techniques

We used a mixed-methods approach to analyze both quantitative and qualitative data, enabling a comprehensive understanding of player behavior and motivations.

✦ Qualitative

Through opt-in user interviews, we explored how players describe their experiences and the value they derive from gameplay. These conversations provided deeper insight into how users interpret cognitive engagement, revealing nuances that are not captured through survey data alone.

✦ Quantitative

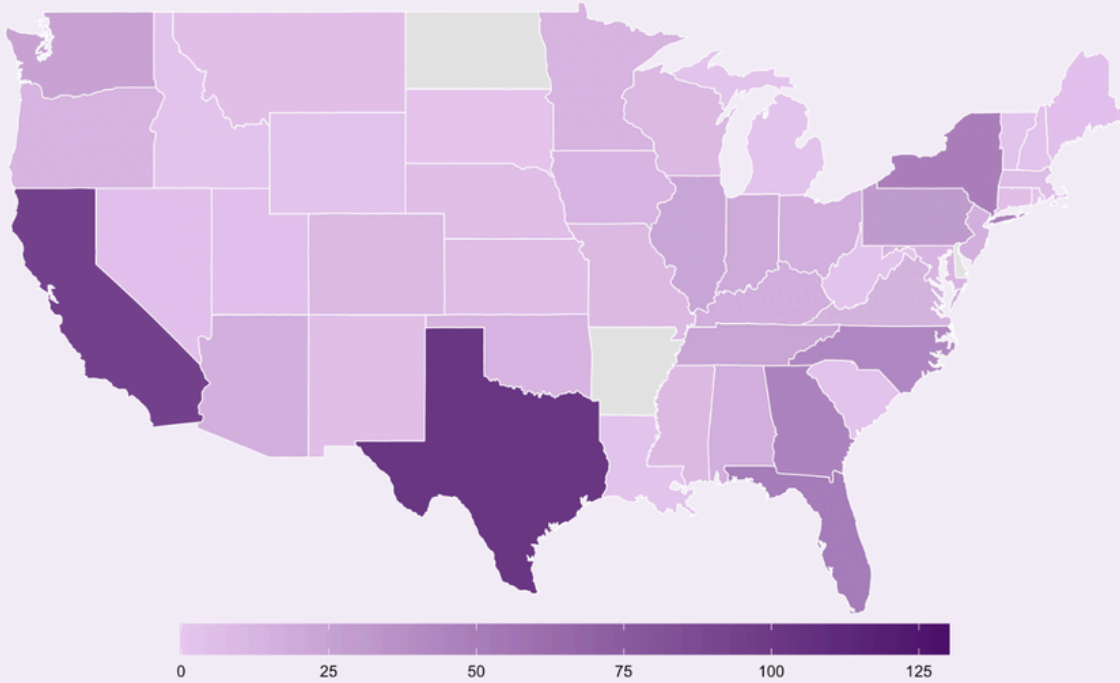
We conducted exploratory data analysis (EDA) on survey responses to identify patterns across demographics and player motivations. This included analyzing distributions of responses and relationships between key variables to understand how different user segments engage with the platform.

Together, this approach allowed us to move beyond surface-level trends and develop a structured segmentation of users based on motivation. These findings form the foundation for the insights and strategic recommendations presented in the following sections.

Customer Discovery: Survey Demographics

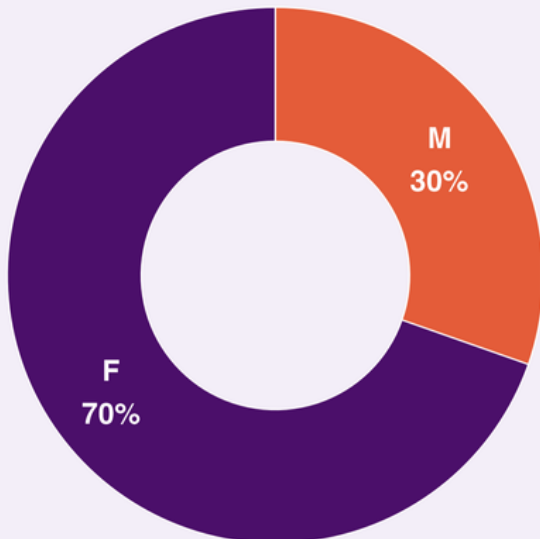
Where Are They Playing From?

Respondents by U.S. state — non-US entries excluded



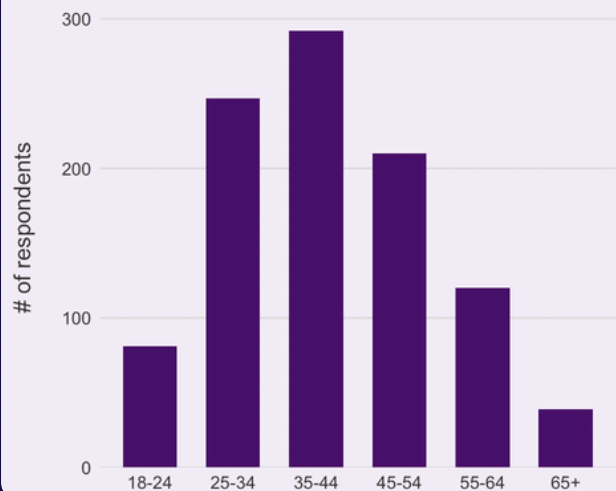
Gender Breakdown

Survey respondents by sex



Who's Playing?

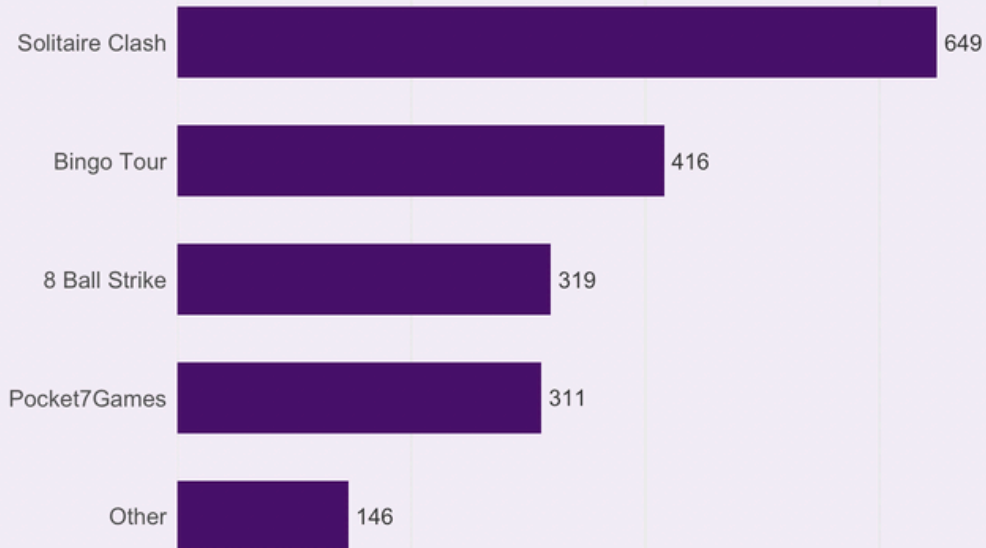
Respondents by age group



Customer Discovery: Games & Motivations

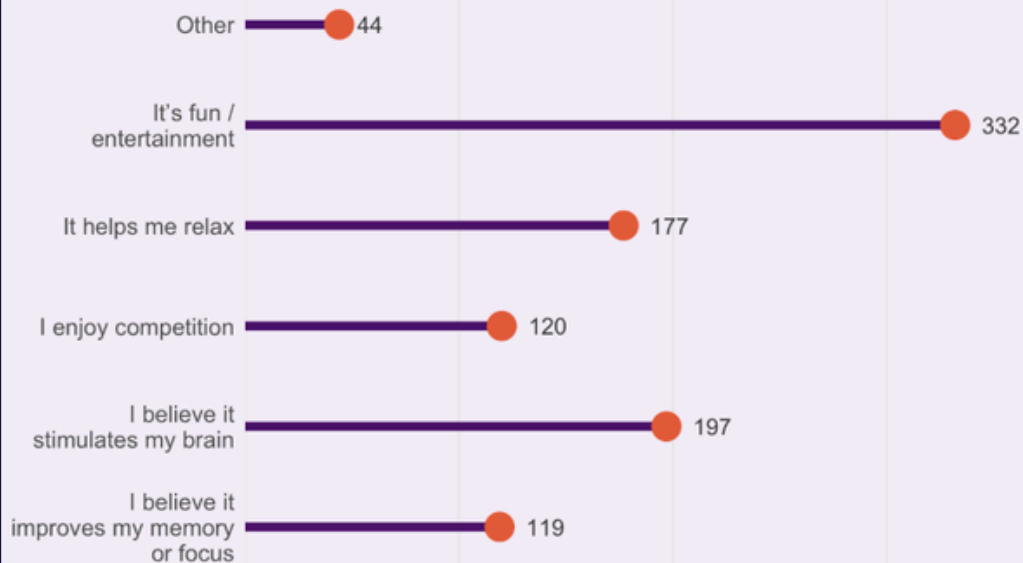
Which Games Are Being Played?

Total mentions per game — respondents could select multiple



Why Do They Play?

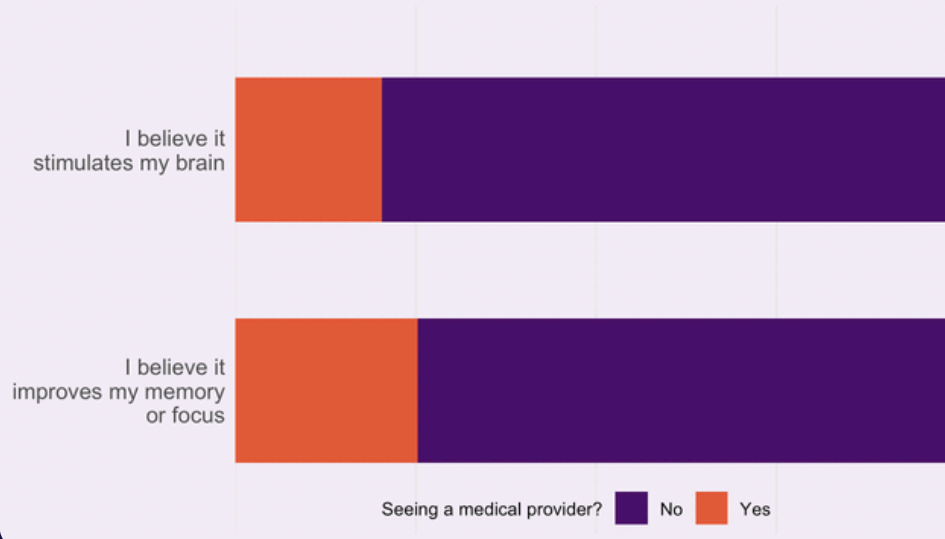
Primary motivation for playing AviaGames products



Customer Discovery: Medical Intersection

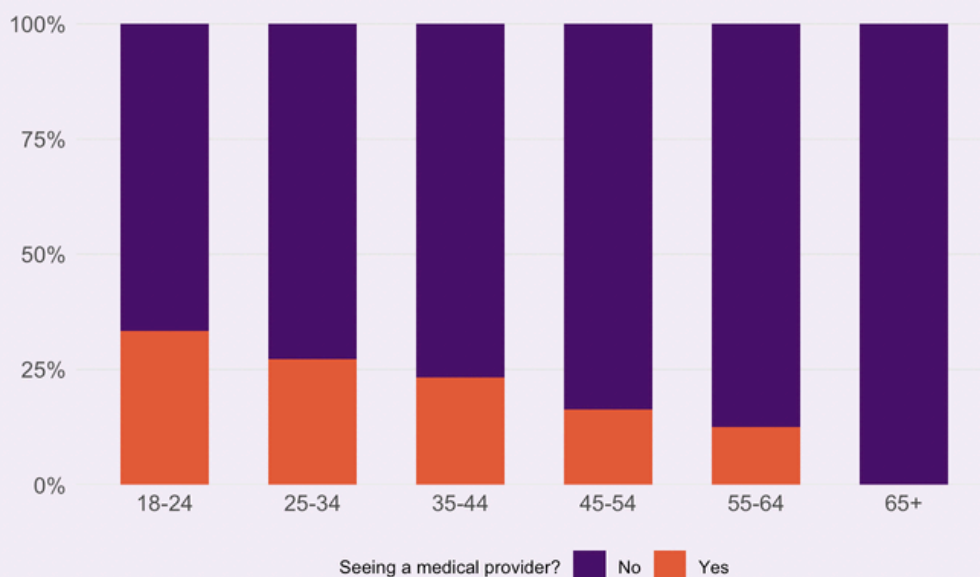
Do Motivations Correlate with Health?

Share of respondents seeing a brain-related medical provider, by motivation



Does Age Correlate with Health?

Share of respondents seeing a brain-related medical provider, by age group



Insights from EDA

1. Avia's player base skews older and predominantly female

Survey data shows a majority female user base (~70%) and strong representation in middle-aged and older demographics. This aligns with patterns seen in casual and routine-based gaming, where users are more likely to value relaxation, consistency, and low-friction engagement.

2. Geographic distribution is broad, with strong U.S. coverage

Players are distributed across the U.S., with higher concentrations in states like Texas and California. This indicates that engagement is not region-specific, suggesting that observed behaviors and motivations are broadly representative rather than localized.

3. Entertainment is the primary driver—but not the only one

While most users report playing for fun and relaxation, a meaningful portion also associate gameplay with cognitive benefits. This suggests that users layer additional meaning onto their experience beyond pure entertainment.

4. Cognitive motivation exists as a secondary layer of value

Cognitive engagement is often not the primary reason users begin playing, but becomes a reinforcing factor over time. Players interpret gameplay as mentally stimulating, which enhances perceived value and may support continued engagement.

5. Cognitive perceptions are non-clinical and self-defined

Most users who associate gameplay with cognitive benefits are not engaging with medical providers. This indicates these perceptions are personal and experiential, rather than grounded in clinical validation.

6. Opportunity lies in perception—not medical positioning

The opportunity is not in entering the clinical "brain training" space, but in leveraging user perceptions of cognitive engagement through product experience and positioning—without making medical claims.

User Segmentation: Segmentation Approach

To translate our findings into actionable product insights, we segmented users based on their primary motivations for gameplay, rather than purely demographic characteristics. This approach allowed us to focus on why users engage, which is more directly tied to product experience and growth opportunities.

From our survey data, two dominant and addressable segments emerged: users primarily motivated by entertainment and those motivated by perceived cognitive engagement (e.g., improving memory, focus, or mental stimulation). Notably, these segments represent a significant portion of the user base, with 33.7% of respondents identifying with entertainment-driven motivations and 30.8% with cognitively motivated drivers.

While additional motivations (such as competition or relaxation) were present, these two segments represented the most significant and strategically relevant groups in terms of both scale and potential impact.

1. Entertainment-Driven Players (33.7%)

These users primarily engage with AviaGames for fun, relaxation, and enjoyment. Gameplay is viewed as a form of leisure, with value derived from entertainment and low-friction engagement.

This segment represents the core user base and aligns closely with Avia's current product positioning.

2. Cognitively Motivated Players (30.8%)

These users associate gameplay with mental stimulation, including beliefs that playing may support memory, focus, or overall brain engagement. While these motivations are not clinically validated, they play a meaningful role in how users perceive the value of gameplay.

Importantly, this segment does not view Avia as a medical or clinical tool, but rather as an activity that *feels* mentally engaging.

Although these segments are distinct in their primary motivations, they are **not mutually exclusive**. Many users who play for entertainment also report experiencing cognitive engagement as a secondary benefit. This overlap suggests that the strongest product opportunity lies not in building separate experiences for each segment, but in developing a shared value proposition that enhances both entertainment and perceived cognitive engagement.

User Segmentation: Implication for Product Strategy

Rather than positioning Avia as a “brain training” product, the opportunity is to design experiences that:

1. Maintain core entertainment value
 2. Subtly reinforce feelings of mental engagement
 3. Appeal to both segments simultaneously
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User Interviews: How Players Describe Their Experiences

Across both cognitive and entertainment segments, users consistently describe similar in-game experiences — centered around engagement, problem-solving, and routine play — even when their stated motivations differ.

1. Engagement is interpreted differently across users

Users experience the same gameplay moments, but frame them as either relaxation or productivity depending on their motivation.

“While playing, I feel relaxed and happy... I don’t think of anything except the game.”

“It turns gameplay into something that feels productive, not just entertaining.”

2. Cognitive value is felt through experience, not clinical improvement

Even cognitively motivated users do not associate gameplay with measurable cognitive gains, but rather with moment-to-moment mental engagement.

“I wouldn’t say it improved my memory — it’s more about figuring things out and adapting quickly.”

3. Strategy and problem-solving drive perceived value

Players value opportunities for planning, pattern recognition, and strategy, which reinforce the perception of “using their brain.”

“The game rewards looking for second-order consequences... I’ve caught myself applying that same thinking to planning my workday.”

Conclusion and Recommendations

Across both cognitive and entertainment segments, players engage with the game in nearly identical ways and derive value from the same moments of strategy, problem-solving, and engagement. Importantly, users do not view the product as a tool for measurable cognitive improvement, but rather as an experience that feels mentally stimulating. The current gameplay experience already delivers perceived cognitive value — not through outcomes, but through engagement. Because this value is already embedded in the core experience, introducing heavier mechanics or structured “training” risks disrupting the casual, low-friction nature of play. Instead, the opportunity lies in making this value more visible to users — particularly those seeking cognitive benefits.

Next steps

#	Recommendations	Priority
1	Surface player progress and improvement Introduce badges, tiers, or achievement systems that reflect skill, consistency, and growth over time. → <i>Aligns with desire to “feel” improvement without changing gameplay</i>	HIGH
2	Add lightweight performance feedback Highlight trends such as streaks, personal bests, or improvement over time. → <i>Makes cognitive value more tangible without formal evaluation</i>	HIGH
3	Preserve low-friction, casual gameplay Ensure new features integrate seamlessly into short, repeatable sessions while maintaining a low-friction experience, avoiding added complexity or mechanics that feel evaluative or test-like. → <i>Maintains the casual, low-friction experience that drives engagement across both user segments</i>	MEDIUM
4	Frame value as engagement, not medical benefit Position the experience as mentally stimulating rather than making explicit claims about memory or cognitive improvement. → <i>Mitigates risk (FTC / Lumosity precedent)</i>	HIGH